

Utkarsh Singh

Full Stack Developer | Generative AI Engineer

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Summary

Final-year Computer Science student specializing in full-stack development and generative AI systems. Experienced in building scalable REST APIs, implementing retrieval-augmented generation pipelines with vector embeddings, and streamlining LLM inference workflows. Strong foundation in MERN architecture, containerized deployment, and production-ready AI applications.

Technical Skills

Languages: JavaScript, Python, C, SQL

Generative AI: LangChain, Hugging Face, Ollama, RAG, Vector Embeddings, LLM APIs

Backend: Node.js, Express.js, FastAPI, Flask, MongoDB, Supabase, REST API

Frontend: React.js, Tailwind CSS, Zustand, GSAP, Bootstrap

DevOps and Infrastructure: Docker, Docker Compose, GitHub Actions, Linux VPS, CI/CD

Tools: Git, Postman, Cloudinary, Vim, Chrome DevTools

CS Fundamentals: Data Structures, Algorithms, OOP, Operating Systems, Computer Networks, DBMS

Experience

Frontend Developer

Sep 2025 – Oct 2025

Freelance Contract, Remote

- Designed and shipped a production-ready React.js + Tailwind CSS frontend for a restaurant client – 5 pages, 10+ reusable components, and full cross-device compatibility. [\[Live Demo\]](#)

Full Stack Web Development Trainee

Jul 2024 – Aug 2024

GRAS tech via BBDU Training Program, Lucknow, India

- Completed a 6-week intensive MERN stack program, delivering 4 projects – 3 React.js frontends and 1 full-stack application – and earned a completion certificate. [\[View Certificate\]](#)

Projects

SiteWhisper: RAG-Powered Chrome Extension for Real-Time Webpage Intelligence

Jan 2026 – Feb 2026

[\[GitHub\]](#)

- Built a Chrome extension enabling natural-language querying of any live webpage using retrieval-augmented generation. Page text is extracted and chunked in-browser, then processed by a Flask backend for semantic search – eliminating the need for a vector database.
- Reduced token consumption by approximately 40% by ranking chunks via cosine similarity and forwarding only the top-relevant context to a Hugging Face model, with gains from sanitization, deduplication, and context-window tuning.
- Secured API access through a first-launch setup flow prompting users to enter their own Hugging Face token, preventing credential hardcoding and enabling user-owned API usage.

CodeSentinel: Automated AI Code Review and Pull Request Analysis System

Jan 2026

[\[GitHub\]](#)

- Engineered a code-review system triggered on every pull request via GitHub Actions, where diffs are processed through a FastAPI backend and LangChain pipeline to generate categorized, prioritized feedback.
- Published AI-generated review comments onto PR threads via the GitHub REST API; severity-based prioritization cut review time by approximately 40% and lowered the risk of merging defective code.
- Containerized the service with Docker and deployed on a Linux VPS via GitHub Actions, integrating AI-assisted review as a seamless step in the pull request lifecycle.

E-Commerce Platform: Full Stack MERN Application

Dec 2025

[\[GitHub\]](#) [\[Live Demo\]](#)

- Architected a secure MERN e-commerce platform with an Admin Dashboard, role-based access control, and JWT authentication with refresh-token rotation for session management.
- Integrated Cloudinary CDN for media delivery and deployed via Docker Compose on a Linux VPS; optimized asset compression cut average image payload by ~50%, improving page load performance.

Education

Babu Banarasi Das University

2022 – Expected May 2026

Bachelor of Technology in Computer Science and Engineering – CGPA: 8.0